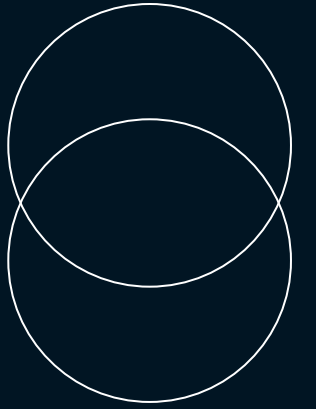
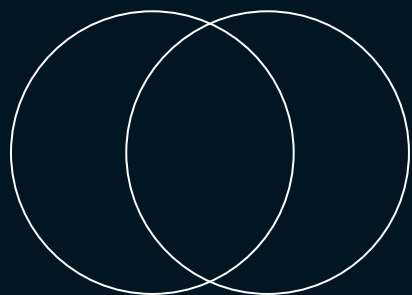




# Final Listing Price Prediction for Private Used Car Sellers

Group 2 - Ji-Soo Kim, Tae-Yoon Kim, Jun-Beom Lee, Jin-Joo Yang, Sung-Hyun Kim



# 0. Recap - Problem statement

## Target Audience



### Used-car owners

those who want to sell their cars as a used car with a reasonable & profitable price



### Used-car Dealer

those who want to afford trust to customers

# 0. Intro - Background research

## Used-car Dealer

Can build trust with customers using reasonable price

## Used-car owners

1. Can predict their used car market price(value) based on sale price prediction

2. Fair Trade Tools for used car owners (Prevent dealer fraud)

kbb.com/faq/values/

### How Do I See The Blue Book® Value For My Vehicle?

To get to the value for your car, navigate the path to the Blue Book Trade-In and Private Party Values:

- On the home page or under "Car Values" from the top navigation, select "My Car's Value".
- Tell us the year, make, model and mileage of the car you own (2015> Honda> Civic>30000 miles). Verify the ZIP code.
- Choose your car's category (sedan vs wagon) and style (DX, EX, LX). Note: Many vehicles only come in one category, but most vehicles are sold in more than one style, which some sites call "trim". Whatever you call it, these are pre-packaged levels of equipment – and they definitely affect the price or value of a car.
- Add any additional equipment or options (packages, alloy wheels, moon roof, premium sound, etc.)
- Choose what you're most likely to do – Trade In to a Dealer or Sell to a Private Party.
- Tell us the car's condition (Excellent, Very Good, Good, or Fair) or, if you're not sure, take the Condition Quiz. Most cars we see are in "Good" condition.
- View the Blue Book Value based on your ZIP code and your car's age, mileage, equipment, and condition.

edmunds.com/car-buying/car-color-facts-and-fictions.html

### SEE EDMUNDS PRICING DATA

## Has Your Car's Value Changed?

Used car values are constantly changing. Edmunds lets you track your vehicle's value over time so you can decide when to sell or trade in.



Feb Apr May Jun

Year Make Model See pricing history

But times may be changing. Color experts predict we're about to become more daring.

So while we know what we want (or think we do) when it comes to the color of the vehicle we're buying, there are lots of questions about colors that most of us consider before making that final decision.

Should we really follow the crowd, or is this the year to break out? Is one color safer than another when it comes to avoiding an accident? Are there colors that are likely to get a car stolen? Is a white car really cooler in the desert? Does a red car beg for a speeding ticket? What about higher insurance premiums for scarlet cars? Is that myth or reality? What color requires the least maintenance?

To separate car color facts from fiction, Edmunds turned to the experts.

# 1. Feature Preprocessing – ‘major options’

## <value counts of each options>

|    | Option                  | Count  |
|----|-------------------------|--------|
| 11 | Bluetooth               | 114149 |
| 12 | Backup Camera           | 113149 |
| 5  | Alloy Wheels            | 107720 |
| 10 | Heated Seats            | 66047  |
| 3  | Navigation System       | 60560  |
| 1  | Sunroof/Moonroof        | 50015  |
| 0  | Leather Seats           | 49957  |
| 18 | Remote Start            | 45689  |
| 7  | Blind Spot Monitoring   | 36890  |
| 14 | CarPlay                 | 28110  |
| 8  | Parking Sensors         | 24677  |
| 44 | Android Auto            | 23317  |
| 6  | Third Row Seating       | 18526  |
| 35 | Steel Wheels            | 16653  |
| 4  | Adaptive Cruise Control | 16598  |
| 9  | Premium Package         | 9890   |
| 19 | Quick Order Package     | 9888   |
| 2  | Power Package           | 7306   |
| 29 | Tow Package             | 6803   |

|    |                            |      |
|----|----------------------------|------|
| 15 | Convenience Package        | 6503 |
| 38 | Multi Zone Climate Control | 5642 |
| 22 | Premium Wheels             | 5605 |
| 21 | Technology Package         | 4939 |
| 23 | Heat Package               | 4497 |
| 51 | Chrome Wheels              | 4091 |
| 24 | Cold Weather Package       | 4001 |
| 43 | Appearance Package         | 3962 |
| 17 | Preferred Package          | 3962 |
| 40 | Suspension Package         | 3622 |
| 20 | Sport Package              | 3363 |
| 13 | SE Package                 | 2939 |
| 37 | Adaptive Suspension        | 2566 |
| 25 | LE Package                 | 2521 |
| 47 | Trailer Package            | 2264 |
| 33 | Memory Package             | 2227 |
| 34 | Comfort Package            | 2043 |
| 54 | Luxury Package             | 1911 |
| 55 | Off Road Package           | 1903 |
| 36 | Driver Assistance Package  | 1566 |

extract only the ‘packages’

| unique_filtered_packages |            |              |                           |
|--------------------------|------------|--------------|---------------------------|
|                          | make_name  | model_name   | filtered_packages         |
| 6                        | Hyundai    | Elantra      | Sport Package             |
| 6                        | Hyundai    | Elantra      | Audio Package             |
| 6                        | Hyundai    | Elantra      | Heat Package              |
| 6                        | Hyundai    | Elantra      | Premium Package           |
| 7                        | Chevrolet  | Malibu       | Driver Confidence Package |
| ...                      | ...        | ...          | ...                       |
| 1573288                  | Volkswagen | Jetta Hybrid | Premium Audio Package     |
| 1573288                  | Volkswagen | Jetta Hybrid | Heat Package              |
| 1573288                  | Volkswagen | Jetta Hybrid | Audio Package             |
| 1573352                  | GMC        | Yukon Hybrid | Comfort Package           |
| 1573352                  | GMC        | Yukon Hybrid | Convenience Package       |

11732 rows × 3 columns

# 1. Feature Preprocessing - ‘major options’

## Binary encoding

```
major_options \
2 [Alloy Wheels, Bluetooth, Backup Camera, Heate...
5 [Leather Seats, Sunroof/Moonroof, Navigation S...
9 [Leather Seats, Navigation System, Adaptive Cr...
10 [Leather Seats]
12 [Sunroof/Moonroof, Alloy Wheels, Bluetooth]
23 [Leather Seats, Navigation System, Adaptive Cr...
36 [Sport Package, Sunroof/Moonroof, Adaptive Cru...
38 [Driver Confidence Package, Power Package, Pre...
40 [Leather Seats, Sunroof/Moonroof, Navigation S...
41 [Power Package, Preferred Package, Third Row S...

binary_encoded
2 000000000000000010000001001000000000000000...
5 000000000000100010000001001000000000000000...
9 000000000000100010000001011000000000000000...
10 0000000000000000000000000000000000000000...
12 0000000000000000100000000100000000000000...
23 000000000000100010000001011000000000000000...
36 0000000000001000000001010110100000000000...
38 00000000000000000001000010110100000010000...
40 0000000000000000100000010010000000000000...
41 0000000000000000001000010110100000010000...
```

## Frequency Counts

```
packages package_score
38 [Driver Confidence Package, Power Package, Pre... 6
40 [] 0
41 [Power Package, Preferred Package, Heat Packag... 5
45 [Power Package, Preferred Package, Technology ... 7
47 [Power Package, Preferred Package, Heat Packag... 6
```

| binary_encoded | feature_1 | feature_2 | feature_3 | feature_4 | feature_5 | ... |
|----------------|-----------|-----------|-----------|-----------|-----------|-----|
| "1010110"      | 1         | 0         | 1         | 0         | 1         | ... |
| "0001011"      | 0         | 0         | 0         | 1         | 0         | ... |

# 1. Feature Preprocessing - 'interior\_color'

## Interior\_color vs listing\_color

| interior_color                                             |        |
|------------------------------------------------------------|--------|
| Black                                                      | 871393 |
| Gray                                                       | 195908 |
| Jet Black                                                  | 186195 |
| Black (Ebony)                                              | 142839 |
| Black (Charcoal)                                           | 112051 |
| ...                                                        |        |
| Brown (Cappuccino w/Heated Lincoln Soft Touch Front Seats) | 1      |
| Circuit Red Nuluxe[nuluxe] With Dark Gray Streamli         | 1      |
| Nut Brown/ Black Leather                                   | 1      |
| Black/Orange w/Fabric Seat Trim (FD)                       | 1      |
| Brown (Espresso/iv/tan/Esp/iv/iv)                          | 1      |

| listing_color |        |
|---------------|--------|
| WHITE         | 666564 |
| BLACK         | 587999 |
| UNKNOWN       | 399905 |
| SILVER        | 384779 |
| GRAY          | 377442 |
| RED           | 252917 |
| BLUE          | 249758 |
| GREEN         | 24074  |
| BROWN         | 22611  |
| ORANGE        | 11631  |
| GOLD          | 10297  |
| TEAL          | 5453   |
| YELLOW        | 5003   |
| PURPLE        | 1468   |
| PINK          | 139    |

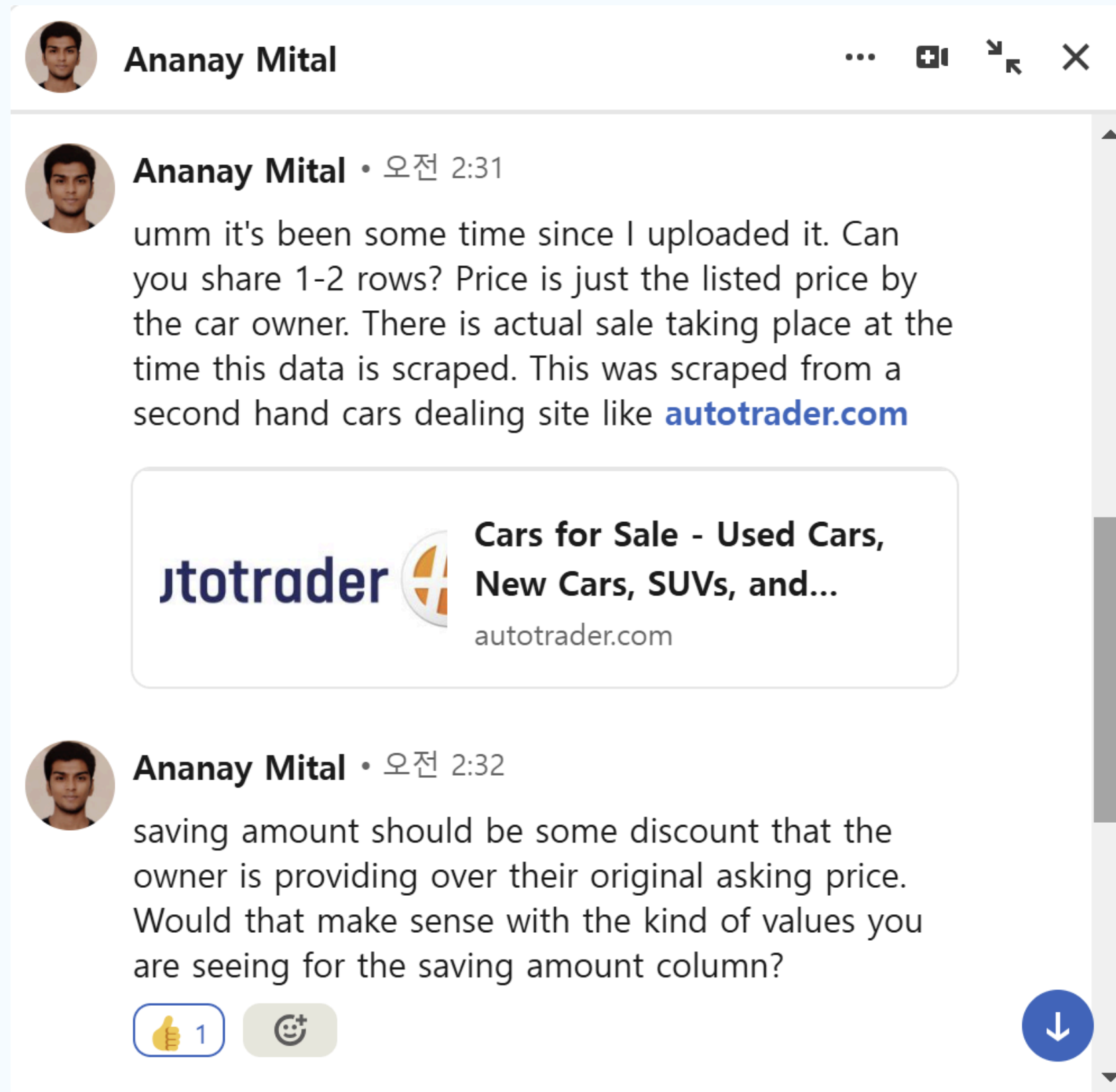
Two Tone colors?

-> process as a separate value

reference?



# 1. Feature Preprocessing - 'price' & 'savings\_amount' ?



**Price** : Price 'posted' on used car site (sold price X)

**Savings\_amount** : A reduced price (if the price is lowered) compared to the original price that was initially posted

# 1. Feature Preprocessing - 'price' & 'savings\_amount' ?

|         | price | savings_amount | price2 |
|---------|-------|----------------|--------|
| 38      | 14639 | 1749           | 12890  |
| 40      | 32000 | 1861           | 30139  |
| 41      | 23723 | 3500           | 20223  |
| 45      | 22422 | 2416           | 20006  |
| 47      | 29424 | 2254           | 27170  |
| ...     | ...   | ...            | ...    |
| 3000033 | 5371  | 453            | 4918   |
| 3000034 | 40993 | 2220           | 38773  |
| 3000035 | 17998 | 381            | 17617  |
| 3000038 | 26998 | 849            | 26149  |
| 3000039 | 19900 | 1203           | 18697  |

[1094520 rows x 3 columns]

**price2 = price - savings\_amount**

Replace 'price' feature with price2  
=> so that the price can be predicted as close  
to the '**selling price**' as possible.

## 2. Our specific topic?



Ananay Mital · 오후 5:22

1. Depends on the website to put this data out for the public.
2. Such a data doesn't help neither a buyer or a seller. Insights from such data is useful through a machine learning model but raw data isn't so no point for any website to show it



Ananay Mital · 오후 5:24

If you have the scraper code. You can identify the potentially sold cars based on the listing ID but found on day  $n+1$  compared to day  $n$



Ananay Mital · 오후 5:25

but then again it would be potentially sold because people might delete listings and/or repost



Using scraping code : Can check whether a specific vehicle existed on a specific date ( $n$  days) based on the list ID, and if the ID disappears on the next day ( $n+1$  days), the vehicle is likely to have been sold.

However, since people can simply delete or re-upload a vehicle listing, **we cannot be sure that it was sold** simply because the listID disappears.

=> **Hard to figure out whether the car is sold or not...** 😭

## 2. Our specific topic?

# ‘Final listed’ Price Prediction of Private Sales Vehicles with Used Car Datas

Because the ‘price’ feature is not the “**sold**” price  
it’s the “**listed price**” on the website...



### 3. Data preprocessing – Missing value

|                         |            |                      |          |
|-------------------------|------------|----------------------|----------|
| vehicle_damage_category | 100.000000 | wheel_system         | 4.891001 |
| combine_fuel_economy    | 100.000000 | mileage              | 4.812836 |
| is_certified            | 100.000000 | trim_name            | 3.876415 |
| bed                     | 99.347742  | trimId               | 3.860849 |
| cabin                   | 97.882262  | engine_type          | 3.352655 |
| is_oemcpo               | 95.487993  | engine_cylinders     | 3.352655 |
| is_cpo                  | 93.903481  | fuel_type            | 2.757430 |
| bed_height              | 85.696924  | description          | 2.596665 |
| bed_length              | 85.696924  | transmission         | 2.139471 |
| owner_count             | 50.566426  | transmission_display | 2.139471 |
| fleet                   | 47.552533  | exterior_color       | 1.665144 |
| theft_title             | 47.552533  | seller_rating        | 1.362382 |
| isCab                   | 47.552533  | body_type            | 0.451427 |
| has_accidents           | 47.552533  | sp_id                | 0.003200 |
| frame_damaged           | 47.552533  | sp_name              | 0.000000 |
| salvage                 | 47.552533  | vin                  | 0.000000 |
| franchise_make          | 19.087579  | savings_amount       | 0.000000 |
| torque                  | 17.259537  | price                | 0.000000 |
| highway_fuel_economy    | 16.375948  | model_name           | 0.000000 |
| city_fuel_economy       | 16.375948  | make_name            | 0.000000 |
| power                   | 16.047319  | longitude            | 0.000000 |
| interior_color          | 12.799363  | listing_id           | 0.000000 |
| main_picture_url        | 12.302936  | listing_color        | 0.000000 |
| major_options           | 6.668178   | listed_date          | 0.000000 |
| engine_displacement     | 5.746123   | latitude             | 0.000000 |
| horsepower              | 5.746123   | is_new               | 0.000000 |
| back_legroom            | 5.308896   | franchise_dealer     | 0.000000 |
| wheelbase               | 5.308896   | dealer_zip           | 0.000000 |
| maximum_seating         | 5.308896   | daysonmarket         | 0.000000 |
| width                   | 5.308896   | city                 | 0.000000 |
| length                  | 5.308896   | year                 | 0.000000 |
| height                  | 5.308896   |                      |          |
| front_legroom           | 5.308896   |                      |          |
| fuel_tank_volume        | 5.308896   |                      |          |
| wheel_system_display    | 4.891001   | dtype: float64       |          |

Missing value ratio of  
each features

85% up : delete

85% down : consider separately

### 3. Data preprocessing - Missing value

|                      |           |
|----------------------|-----------|
| bed_height           | 85.696924 |
| bed_length           | 85.696924 |
| owner_count          | 50.566426 |
| fleet                | 47.552533 |
| theft_title          | 47.552533 |
| isCab                | 47.552533 |
| has_accidents        | 47.552533 |
| frame_damaged        | 47.552533 |
| salvage              | 47.552533 |
| franchise_make       | 19.087579 |
| torque               | 17.259537 |
| highway_fuel_economy | 16.375948 |

same ratio of missing values

```
columns_to_check = ['fleet', 'theft_title', 'isCab', 'has_accidents', 'frame_damaged', 'salvage']

# extract the indexes of the data samples including missing values of each features
missing_indices = [set(df[df[col].isna()].index) for col in columns_to_check]

# check whether the rows including missing values are all same among all features
all_equal = all(missing_indices[0] == indices for indices in missing_indices)

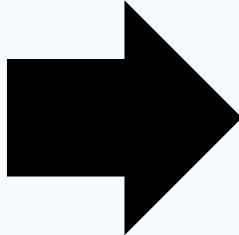
# outcome
if all_equal:
    print("Rows including missing values are all the same.")
else:
    print("Rows including missing values are not same.")
```

Rows including missing values are all the same.

### 3. Data preprocessing - Missing value

|                      | Missing Values | Percentage (%) |
|----------------------|----------------|----------------|
| city_fuel_economy    | 246997         | 15.697848      |
| highway_fuel_economy | 246997         | 15.697848      |
| interior_color       | 243832         | 15.496697      |
| torque               | 201806         | 12.825742      |
| owner_count          | 90418          | 5.746499       |
| major_options        | 70971          | 4.510549       |
| maximum_seating      | 67164          | 4.268595       |
| engine_displacement  | 63925          | 4.062741       |
| horsepower           | 63925          | 4.062741       |
| wheel_system         | 50942          | 3.237609       |
| trim_name            | 44536          | 2.830477       |
| exterior_color       | 35323          | 2.244947       |
| fuel_type            | 32573          | 2.070171       |
| transmission         | 26791          | 1.702697       |
| mileage              | 19390          | 1.232328       |
| body_type            | 1862           | 0.118339       |
| price                | 0              | 0.000000       |
| salvage              | 0              | 0.000000       |
| model_name           | 0              | 0.000000       |
| theft_title          | 0              | 0.000000       |
| savings_amount       | 0              | 0.000000       |
| listed_date          | 0              | 0.000000       |
| make_name            | 0              | 0.000000       |
| listing_color        | 0              | 0.000000       |
| city                 | 0              | 0.000000       |
| isCab                | 0              | 0.000000       |
| has_accidents        | 0              | 0.000000       |
| franchise_dealer     | 0              | 0.000000       |
| frame_damaged        | 0              | 0.000000       |
| fleet                | 0              | 0.000000       |
| engine_type          | 0              | 0.000000       |
| dealer_zip           | 0              | 0.000000       |
| daysonmarket         | 0              | 0.000000       |
| year                 | 0              | 0.000000       |

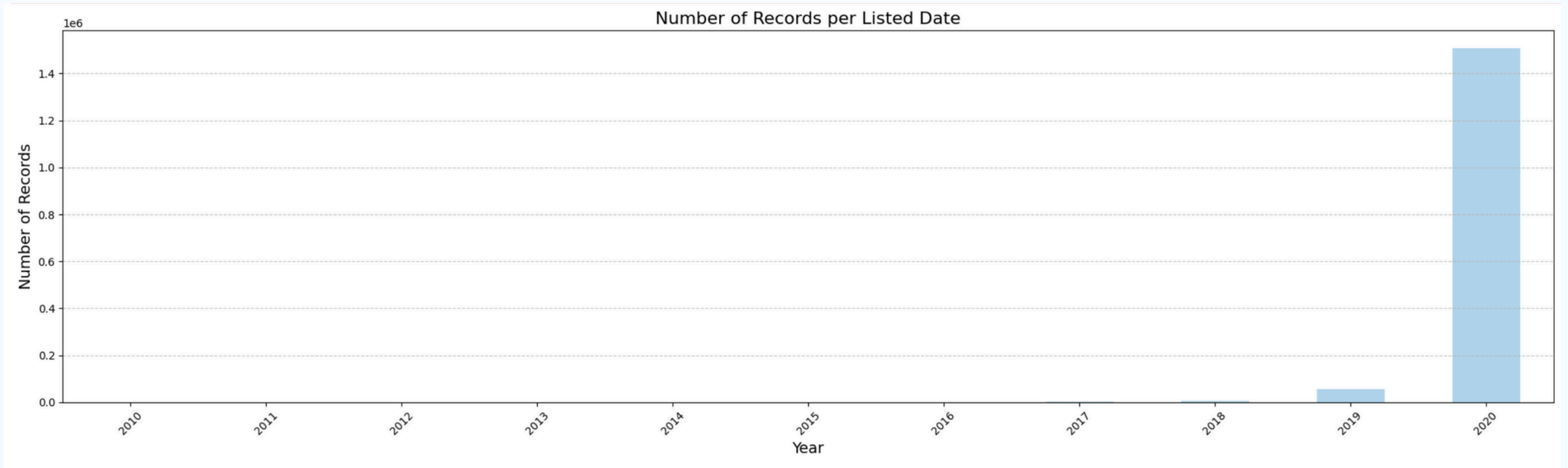
1,500,000 data samples



|                      | Missing Values | Percentage (%) |
|----------------------|----------------|----------------|
| interior_color       | 192683         | 14.526238      |
| torque               | 129741         | 9.781085       |
| owner_count          | 71963          | 5.425241       |
| major_options        | 46695          | 3.520304       |
| exterior_color       | 26862          | 2.025108       |
| mileage              | 14944          | 1.126618       |
| maximum_seating      | 10500          | 0.791588       |
| transmission         | 10372          | 0.781938       |
| engine_displacement  | 8488           | 0.639904       |
| horsepower           | 8488           | 0.639904       |
| wheel_system         | 818            | 0.061668       |
| fuel_type            | 16             | 0.001206       |
| body_type            | 9              | 0.000679       |
| price                | 0              | 0.000000       |
| make_name            | 0              | 0.000000       |
| savings_amount       | 0              | 0.000000       |
| model_name           | 0              | 0.000000       |
| theft_title          | 0              | 0.000000       |
| trim_name            | 0              | 0.000000       |
| salvage              | 0              | 0.000000       |
| listed_date          | 0              | 0.000000       |
| listing_color        | 0              | 0.000000       |
| city                 | 0              | 0.000000       |
| isCab                | 0              | 0.000000       |
| highway_fuel_economy | 0              | 0.000000       |
| has_accidents        | 0              | 0.000000       |
| franchise_dealer     | 0              | 0.000000       |
| frame_damaged        | 0              | 0.000000       |
| fleet                | 0              | 0.000000       |
| engine_type          | 0              | 0.000000       |
| dealer_zip           | 0              | 0.000000       |
| daysonmarket         | 0              | 0.000000       |
| city_fuel_economy    | 0              | 0.000000       |
| year                 | 0              | 0.000000       |

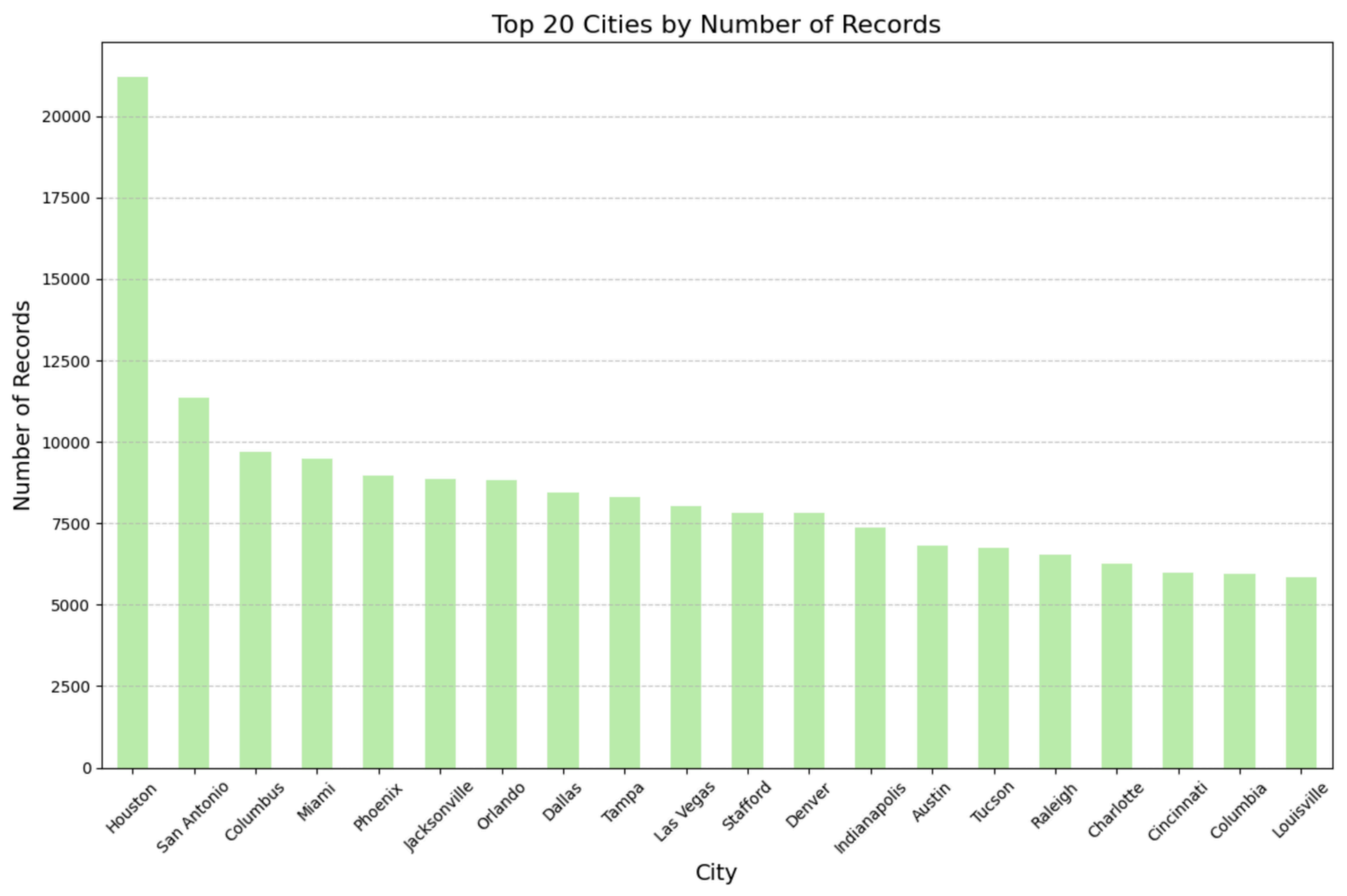
1,300,000 data samples

### 3. Data preprocessing - Listed Date per Year



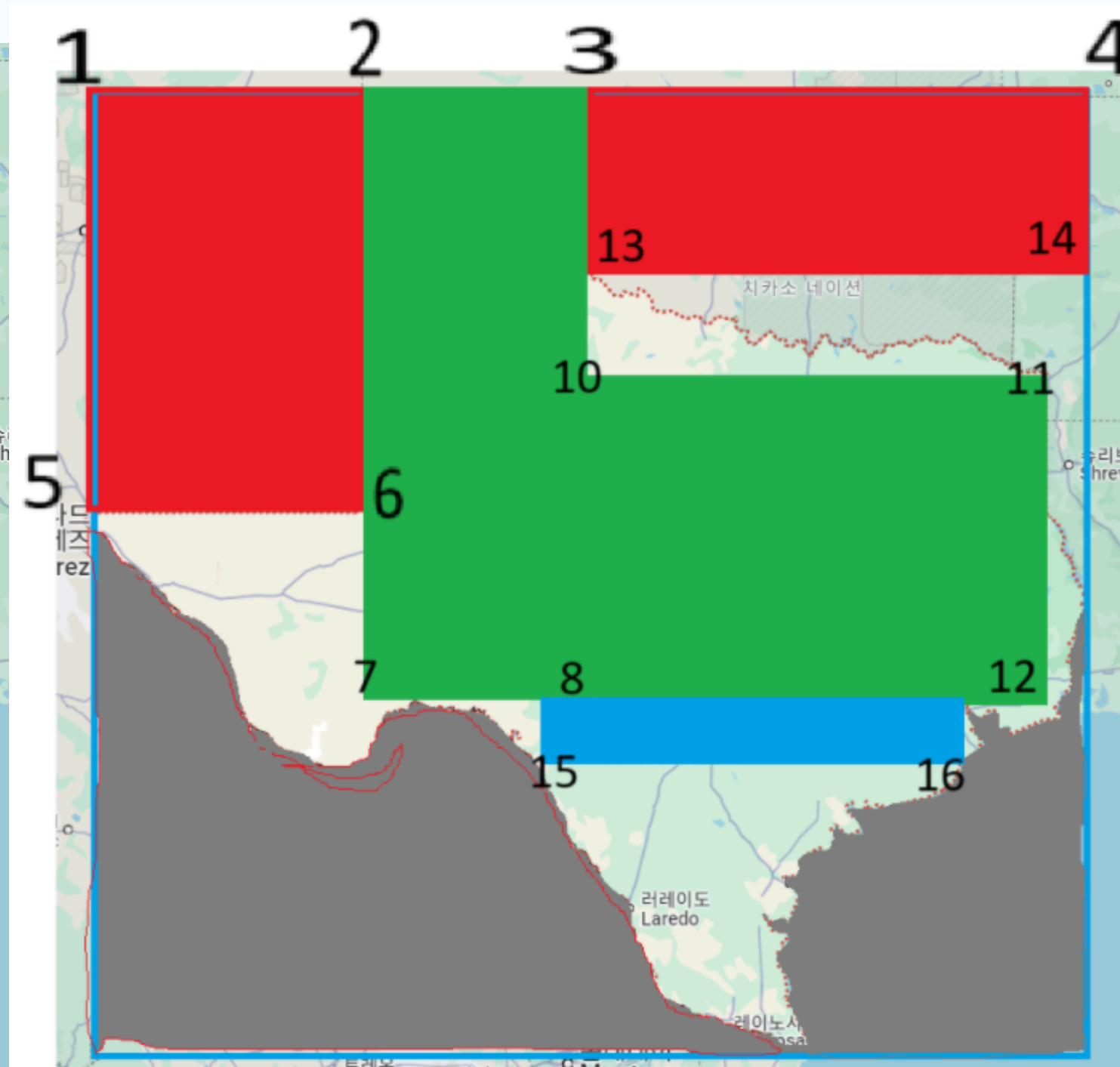
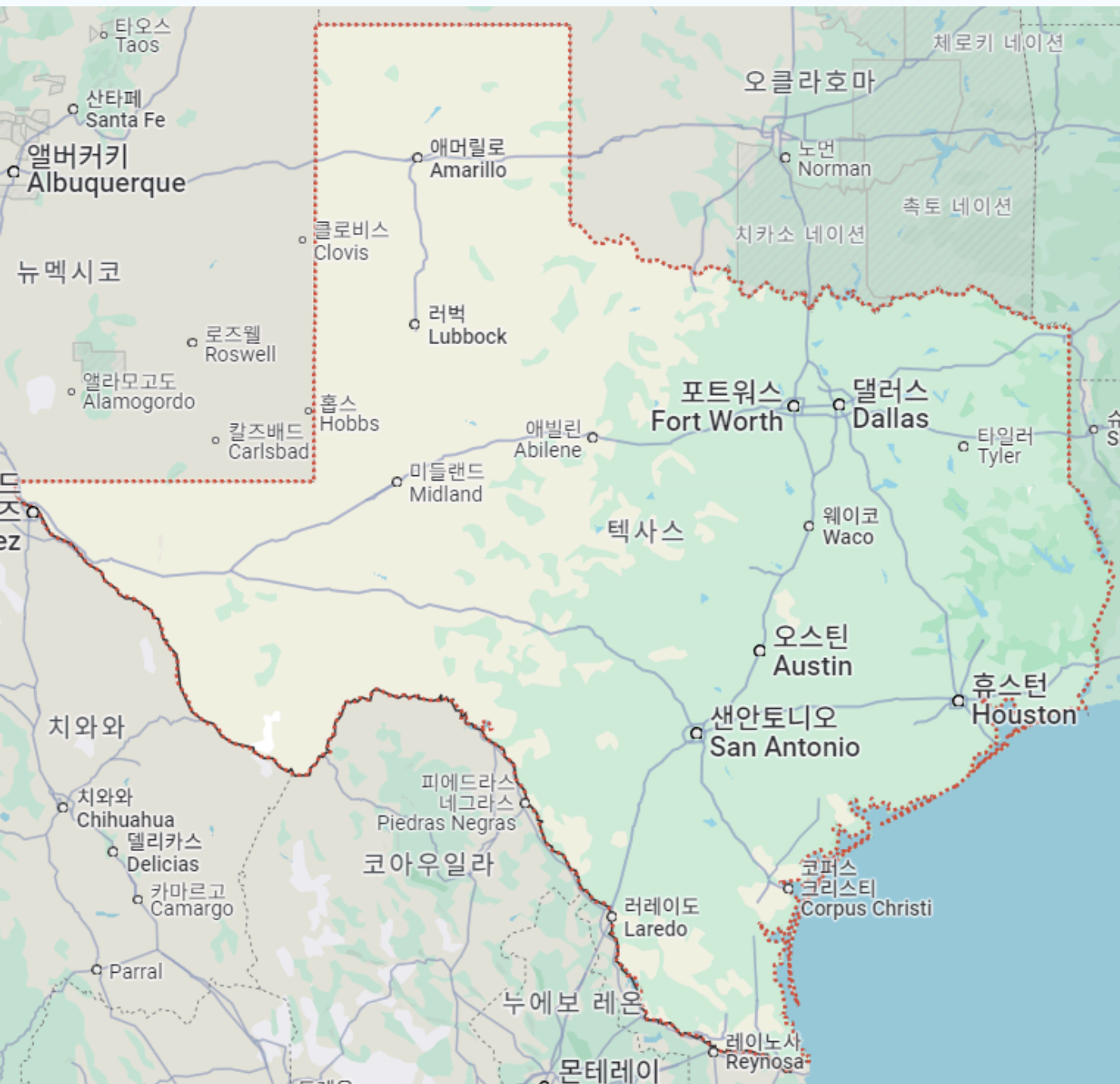
**Our range of Year :**  
**3-9-2019 ~ 2-9-2020**

### 3. Data preprocessing - Top 20 Cities



### 3. Data preprocessing - city : “Texas”

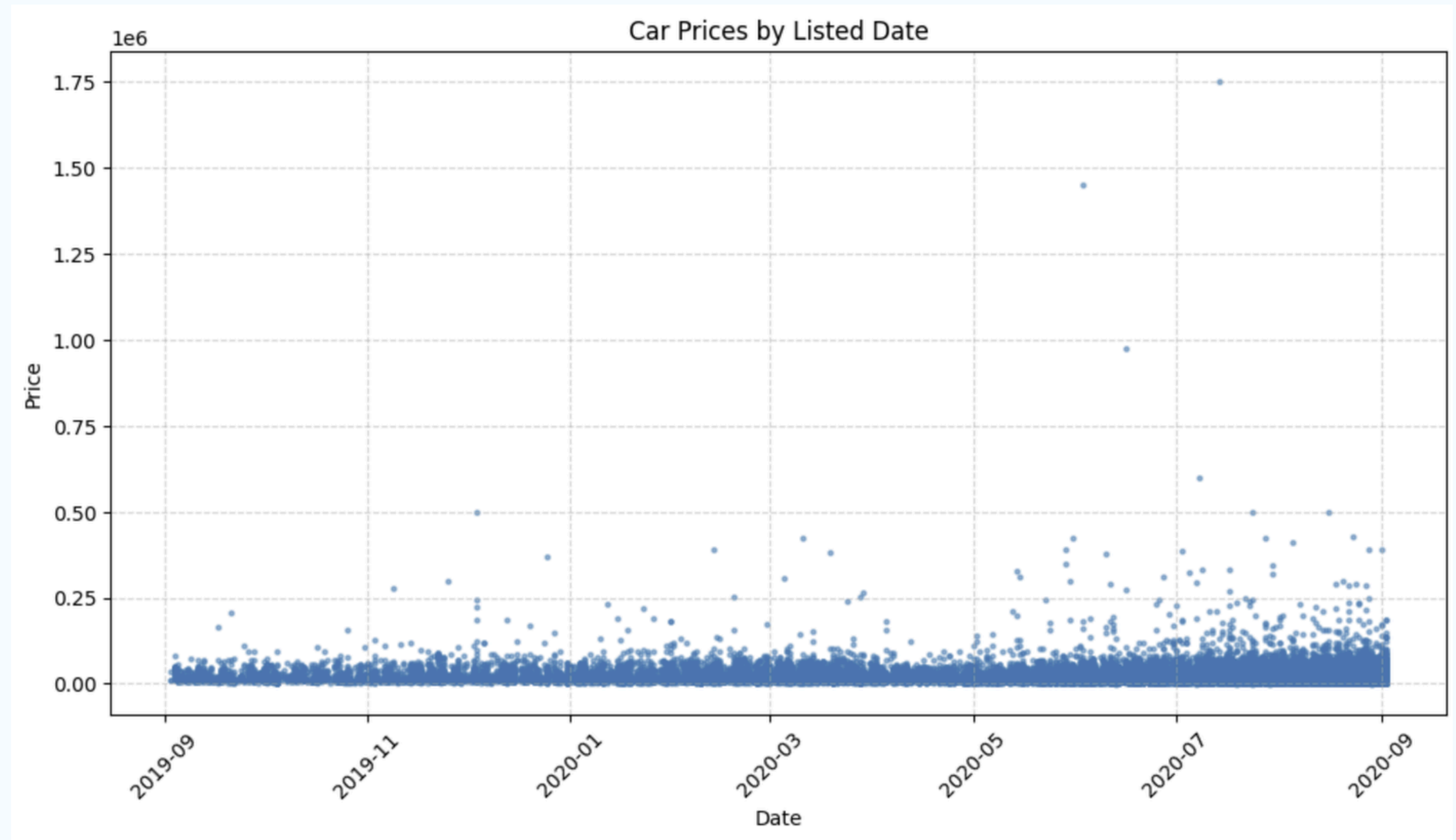
(latitude, longitude)



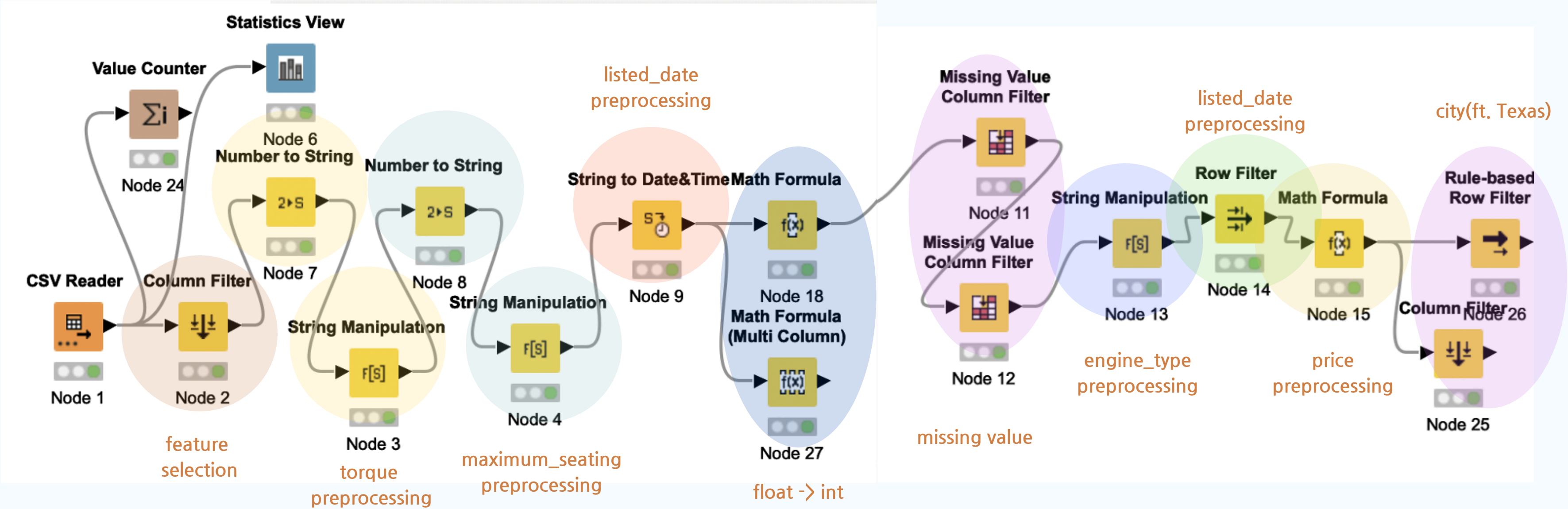
- 1: (36.49, -106.59)
- 2: (36.49, -103.02)
- 3: (36.49, -100.01)
- 4: (36.49, -93.5)
- 5: (32, -106.59)
- 6: (32, -103.05)
- 7: (29.88, -103.05)
- 8: (29.88, -100.01)
- 10: ( 33.55, -100.01)
- 11: (33.55, -94.05)
- 12: (29.88, -94.05)
- 13: (34.58, -100)
- 14: (34.58, -93.5)
- 15: (29.2, -100.01)
- 16: (29.2, -95.2)

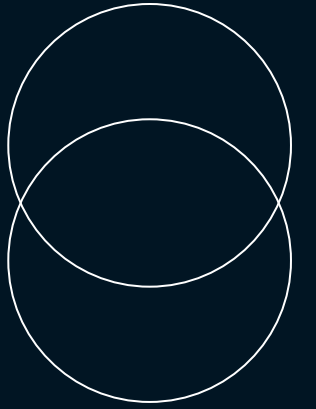
### 3. Data preprocessing - city : “Texas”

|    | A        | B         | C            | D     |
|----|----------|-----------|--------------|-------|
| 1  | latitude | longitude | city         | texas |
| 2  | 35.1901  | -106.659  | Albuquerque  | 0     |
| 3  | 27.7626  | -98.048   | Alice        | 1     |
| 4  | 29.1903  | -95.4048  | Angleton     | 1     |
| 5  | 27.9369  | -97.1482  | Aransas Pass | 1     |
| 6  | 34.1906  | -97.1593  | Ardmore      | 0     |
| 7  | 34.3689  | -96.1389  | Atoka        | 0     |
| 8  | 29.5065  | -94.9838  | Bacliff      | 1     |
| 9  | 28.9902  | -95.9268  | Bay City     | 1     |
| 10 | 29.7766  | -94.9634  | Baytown      | 1     |
| 11 | 28.3975  | -97.7467  | Beeville     | 1     |
| 12 | 32.6984  | -93.7398  | Benton       | 0     |
| 13 | 33.5933  | -96.1774  | Bonham       | 1     |
| 14 | 34.8519  | -106.691  | Bosque Farms | 0     |
| 15 | 32.5657  | -93.7288  | Bossier City | 0     |
| 16 | 33.5862  | -97.914   | Bowie        | 1     |
| 17 | 34.0115  | -94.7396  | Broken Bow   | 0     |
| 18 | 25.9952  | -97.4846  | Brownsville  | 1     |
| 19 | 34.078   | -98.5563  | Burkburnett  | 1     |
| 20 | 31.8961  | -106.599  | Canutillo    | 1     |



# 4. Brief KNIME Workflow





# Q & A

Thank you for listening!

# Appendix

